

# Database Systems and Information Integration

Autman:2025

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Business Intelligence & Data Science  
International School of Management Hamburg  
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# Outline

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

# Purpose and Aim of the Course

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

## Objective and Scope

2

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Introduction to Database Management Systems (DBMS)
- Relational Databases (SQL)
- Data Modeling and Operational Integration (Normal Forms)
- Modern Approaches (NoSQL)

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

3

### Introduction

### Introduction to DBMS

### Types of Databases

### Database Languages

### Basics of Relational Databases

### Hands-On Tutorial on DQL

### Database Design

### Entity Relationship Model

### Joins, Unions, and String Functions

### Normalization

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## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

4

#### Introduction

#### Introduction to DBMS

#### Types of Databases

#### Database Languages

#### Basics of Relational Databases

#### Hands-On Tutorial on DQL

#### Database Design

#### Entity Relationship Model

#### Joins, Unions, and String Functions

#### Normalization

### ● Core Textbooks:

- ▶ Silberschatz, Korth & Sudarshan (2019): Database System Concepts. McGraw-Hill Education.
- ▶ Elvis C. Foster, Shripad V. Godbole (2017) - Database Systems- A Pragmatic Approach
- ▶ Kleinschmidt, P., Rank, C. (2017): Relationale Datenbanksysteme Eine praktische Einföhrung. Springer.

## Data Management

| Modul-Nr. / Code          |                                                      | Semester    | Dauer                   |                    | ECTS-Credits        |     |
|---------------------------|------------------------------------------------------|-------------|-------------------------|--------------------|---------------------|-----|
| BEMA2-BIDS1               |                                                      | 2           | 1 Semester              |                    | 6                   |     |
| Art der Lehrveranstaltung |                                                      | Lehrsprache | Häufigkeit des Angebots |                    | SWS                 |     |
| Wahlpflicht               |                                                      | Deutsch     | Sommersemester          |                    | 4                   |     |
| 1                         | Lehrveranstaltungen des Moduls                       |             | Kontaktzeit (h)         | Selbst-studium (h) | Gesamt-workload (h) | SWS |
|                           | Introduction to Data Science & Business Intelligence |             | 15                      | 60                 | 75                  | 2   |
|                           | Database Systems & Information Integration           |             | 15                      | 60                 | 75                  | 2   |

**Type and duration of the examination:** Written exam (120 min).

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

## Objective and Scope

## Introduction

## Introduction to DBMS

## Types of Databases

## Database Languages

## Basics of Relational Databases

## Hands-On Tutorial on DQL

## Database Design

## Entity Relationship Model

## Joins, Unions, and String Functions

## Normalization

5

- Voluntary **In-Class-Presentations** can be held.
- **Task:** Prepare a **15 min PowerPoint-Slide** to discuss **“Modern Approaches in Database Systems”**.
- As base reading choose one of the following **chapters** from the **Silberschatz database book**:
  - ACID vs. BASE / CAP Theorem
  - NoSQL Databases
  - Chapter 10: **Big Data**
  - Data Warehouses vs. Data Lakes
  - Chapter 11: **Data Analytics**
  - Chapter 26: **Blockchain Databases** (including excursus in Crypto Currencies)
  - MySQL Master-Slave-concept (additional)
- Presentations will be graded according to the typical (of a hundred) scale
- If the grade of the presentation exceeds the final exam grade, a 30% bonus including the better presentation grade will be provided, i.e.
- **Final Grade = max( 30% x Presentation Grade + 70% x Written Exam Grade, 100% x Written Exam Grade)**
- Note: You cannot be downgraded by investing time in a presentation.

# About Me – The Serious (and Not-So-Serious) Version

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

### Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

6



- Rushikesh
- Born in western ghats of India.
- IIT Madras alumnus
  - ▶ just lost part of my soul and hair over there
- Worked at TU Dresden and collaborated with MIT, Yale, UC Berkeley and UT Austin
  - ▶ Cool!! Isn't it?
- Lifelong learner
- Beyond the Nerd Life: I love history and culture, travel, lift weights, ride bikes, and burn things in the kitchen - sometimes intentionally

"Life is way beyond our professions"



# Welcome to the class!

## Agenda

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

#### Introduction

7

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- What to expect?
- Interactive Classes
- Hands-on
- Handling real-world problem
- Mini Group Discussions
- Enjoying the learning process

# Database Management Systems (DBMS)

Why to learn it?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

8

- 90% of business data lives in databases
- Today almost every organization stores its core data in a database
- If you cannot access, query, or understand the structure of this data  
→ you cannot analyze it.
- DBMS provides:
  - data integrity
  - consistency
  - constraints
  - controlled access
  - validated input

**“Without understanding databases, you cannot do real analytics.”**

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Data Base & File System  
Brief overview of DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

9

122

# Difference between Data Base & File System

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

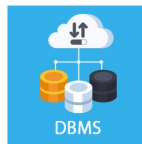
10

- **File systems** are designed to store data in a flat approach

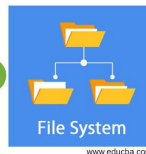
- ▶ Mostly unstructured
- ▶ Difficult to handle the data and find the needed information
- ▶ Easy for single user

### Limitations:

- ▶ **Data Redundancy:** Duplicate entries across files
- ▶ **Inconsistency:** Conflicting or outdated information
- ▶ **Difficult Access:** Manual file search required
- ▶ **Poor Security:** No control over data access
- ▶ **Single-User Access:** No support for collaboration
- ▶ **No Backup/Recovery:** Data loss was often permanent
- ▶ Inherent rawness of data logging



VS



122

# Difference between Data Base & File System

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and Scope

Introduction

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

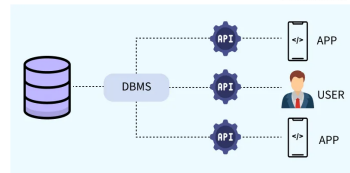
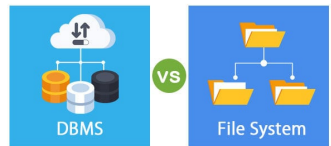
Normalization

11

- **Databases:** collection of well-organized related records in coherent manner

- **DBMS:**

- ▶ Are designed to store data in a structured way
- ▶ Different pieces of data are linked together
- ▶ Typically organized around data models
- ▶ Typically designed to be accessed by multiple users simultaneously
- ▶ Have locking and concurrency control to prevent data corruption



122

# Difference between Data Base & File System

Examples: vehicle tracking data

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

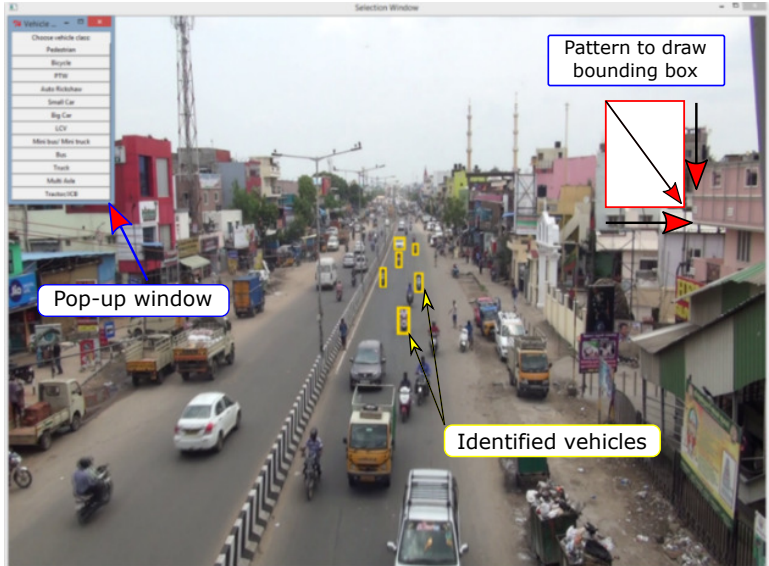
Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

12



122

# Difference between Data Base & File System

Examples: vehicle tracking data

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

## Objective and Scope

## Introduction

## Introduction to DBMS

## Types of Databases

## Database Languages

## Basics of Relational Databases

## Hands-On Tutorial on DQL

## Database Design

## Entity Relationship Model

## Joins, Unions, and String Functions

## Normalization

|     | A | B   | C  | D     | E       | F     | G       | H     | I        |
|-----|---|-----|----|-------|---------|-------|---------|-------|----------|
| 1   |   |     |    |       | point 1 |       | point 2 |       | centroid |
| 3   | 4 | 1-1 | 47 | 0.510 | 39.720  | 2.612 | 24.047  | 1.561 | 31.883   |
| 11  | 4 | 1-1 | 48 | 0.510 | 39.720  | 2.622 | 23.777  | 1.566 | 31.749   |
| 19  | 4 | 1-1 | 49 | 0.472 | 39.312  | 2.583 | 23.504  | 1.528 | 31.408   |
| 27  | 4 | 1-1 | 50 | 0.494 | 38.915  | 2.641 | 23.243  | 1.567 | 31.079   |
| 35  | 4 | 1-1 | 51 | 0.457 | 38.515  | 2.602 | 22.974  | 1.530 | 30.745   |
| 43  | 4 | 1-1 | 52 | 0.478 | 38.125  | 2.612 | 22.713  | 1.545 | 30.419   |
| 51  | 4 | 1-1 | 53 | 0.499 | 37.739  | 2.564 | 22.707  | 1.532 | 30.223   |
| 59  | 4 | 1-1 | 54 | 0.463 | 37.349  | 2.583 | 22.190  | 1.523 | 29.770   |
| 67  | 4 | 1-1 | 55 | 0.463 | 37.349  | 2.602 | 21.679  | 1.532 | 29.514   |
| 75  | 4 | 1-1 | 56 | 0.484 | 36.970  | 2.658 | 21.432  | 1.571 | 29.201   |
| 83  | 4 | 1-1 | 57 | 0.447 | 36.587  | 2.621 | 21.177  | 1.534 | 28.882   |
| 91  | 4 | 1-1 | 58 | 0.468 | 36.215  | 2.630 | 20.928  | 1.549 | 28.572   |
| 99  | 4 | 1-1 | 59 | 0.433 | 35.839  | 2.602 | 20.431  | 1.517 | 28.135   |
| 107 | 4 | 1-1 | 60 | 0.509 | 35.480  | 2.657 | 20.193  | 1.583 | 27.837   |
| 115 | 4 | 1-1 | 61 | 0.474 | 35.111  | 2.620 | 19.946  | 1.547 | 27.529   |
| 123 | 4 | 1-1 | 62 | 0.439 | 34.744  | 2.638 | 19.468  | 1.538 | 27.106   |
| 131 | 4 | 1-1 | 63 | 0.514 | 34.395  | 2.647 | 19.232  | 1.581 | 26.813   |
| 139 | 4 | 1-1 | 64 | 0.404 | 34.381  | 2.656 | 18.997  | 1.530 | 26.689   |
| 147 | 4 | 1-1 | 65 | 0.444 | 33.678  | 2.620 | 18.759  | 1.532 | 26.219   |
| 155 | 4 | 1-1 | 66 | 0.444 | 33.678  | 2.682 | 18.302  | 1.563 | 25.990   |
| 163 | 4 | 1-1 | 67 | 0.410 | 33.324  | 2.646 | 18.069  | 1.528 | 25.697   |
| 171 | 4 | 1-1 | 68 | 0.484 | 32.987  | 2.699 | 17.847  | 1.592 | 25.417   |
| 179 | 4 | 1-1 | 69 | 0.450 | 32.639  | 2.619 | 17.613  | 1.535 | 25.126   |
| 187 | 4 | 1-1 | 70 | 0.416 | 32.294  | 2.681 | 17.172  | 1.548 | 24.733   |
| 195 | 4 | 1-1 | 71 | 0.436 | 31.959  | 2.698 | 16.732  | 1.567 | 24.345   |
| 203 | 4 | 1-1 | 72 | 0.382 | 31.952  | 2.662 | 16.510  | 1.522 | 24.231   |
| 211 | 4 | 1-1 | 73 | 0.422 | 31.290  | 2.679 | 16.079  | 1.550 | 23.684   |
| 219 | 4 | 1-1 | 74 | 0.441 | 30.963  | 2.688 | 15.866  | 1.564 | 23.414   |
| 227 | 4 | 1-1 | 75 | 0.408 | 30.632  | 2.696 | 15.654  | 1.552 | 23.143   |
| 235 | 4 | 1-1 | 76 | 0.480 | 30.317  | 2.712 | 15.234  | 1.596 | 22.776   |
| 243 | 4 | 1-1 | 77 | 0.395 | 29.985  | 2.678 | 15.023  | 1.536 | 22.504   |

- File System: One file is generated for each tracked vehicle
- Inherent raw data logging at 0.2 sec interval
- Cameras installed at multiple locations
- Difficult to infer anything

# Difference between Data Base & File System

Examples: vehicle tracking data

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

12

| Global vehicle ID | Frame Number | Time stamp<br>( MM:SS: MS ) | Vehicle Type* | X coordinate<br>(meters) | Y coordinate<br>(meters) |
|-------------------|--------------|-----------------------------|---------------|--------------------------|--------------------------|
| 1                 | 48           | 00:48:00                    | 4             | 31.88                    | 1.56                     |
| 2                 | 48           | 00:48:00                    | 2             | 53.6                     | 3.06                     |
| 3                 | 48           | 00:48:00                    | 6             | 10.82                    | 2.42                     |
| 4                 | 48           | 00:48:00                    | 2             | 20.66                    | 0.65                     |

\* Each numeral in the column define the different vehicle type

- Database: Refined data in Table format
- Easy for all: information cheat sheet (Meta Data) can be provided
- Insights can be obtained



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Examples: vehicle tracking data

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

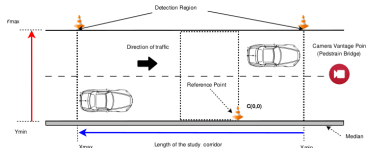
Joins, Unions, and  
String Functions

Normalization

12

Table 1: Detailed information of dataset

| Column Number | Associated Information | Detailed Description                                                                                                                                                                      |
|---------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | Vehicle Type           | 1→Cycle<br>2→Motorcycle<br>3→Auto Rickshaw<br>4→Small Car<br>5→ Big Car<br>6→Light Commercial Vehicle<br>7→ Minibus<br>8→ Bus<br>9→ Truck<br>10→ Multi Axle Truck<br>11→Miscellaneous JCB |
| 2             | Global Vehicle ID      | Consecutive numbers given to vehicles according to the sequence of tracking                                                                                                               |
| 3             | Frame Number           | Processed image frame number<br>Considered time interval is 0.2 seconds                                                                                                                   |
| 3             | X Coordinate           | X co-ordinate of geometric center of vehicle ( <i>meters</i> )                                                                                                                            |
| 4             | Y Coordinate           | Y co-ordinate of geometric center of vehicle ( <i>meters</i> )                                                                                                                            |



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# Difference between Data Base & File System

Examples: vehicle tracking data

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and Scope

Introduction

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

12

| Sales report |         |          |       |       |       |       |   |
|--------------|---------|----------|-------|-------|-------|-------|---|
| Managers     | January | February | March | April | May   | June  |   |
| 1 Aaron      | 20450   | 27298    | 23501 | 27966 | 23709 | 18789 |   |
| 2 Alex       | 0       | 0        | 8445  | 10050 | 12230 | 10583 |   |
| 3 Ashley     | 14017   | 13223    | 16876 | 20082 | 20189 | 15299 |   |
| 4 Blake      | 30892   | 28993    | 34557 | 41123 | 35234 | 29808 |   |
| 5 Caroline   | 27738   | 26775    | 29901 | 35582 | 32830 | 25405 |   |
| 6 Daniel     | 15606   | 14255    | 17118 | 20370 | 16955 | 15341 |   |
|              | 1       | 2        | 3     | 4     | 5     | 6     | 7 |

Records DB

Columns DB

In this database records 6 and 7 columns

Image Source: <https://exceltable.com/en/database-xml/images/database-xml1-1.png>

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# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Data Base & File System  
Brief overview of DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

13

122

# Database Management System

## Overall View

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

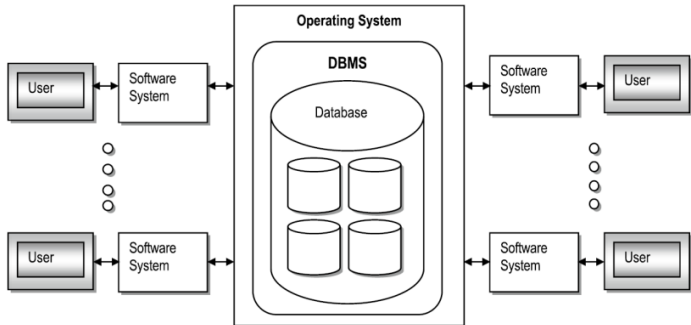
Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

14



# Database Management System Overall View

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

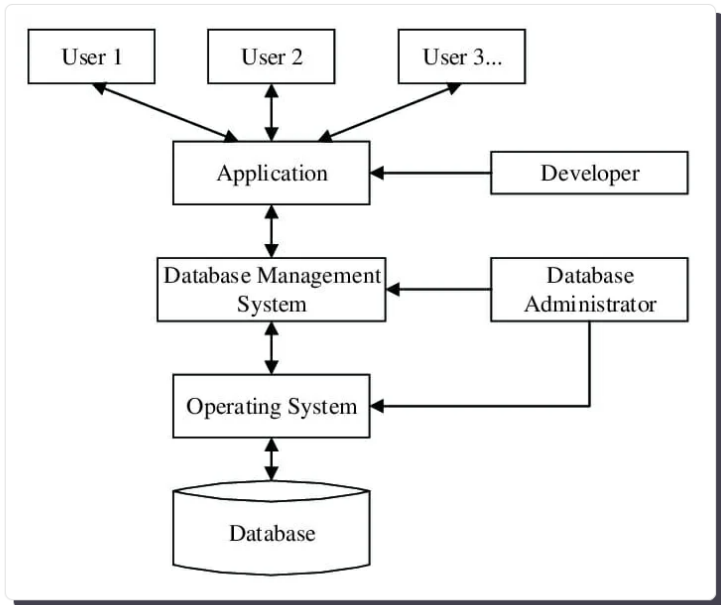
Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

14



# Database Management System

## Steps Involved When Applications Access a Database

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

### Introduction

### Introduction to DBMS

### Types of Databases

### Database Languages

### Basics of Relational Databases

### Hands-On Tutorial on DQL

### Database Design

### Entity Relationship Model

### Joins, Unions, and String Functions

### Normalization

15

- 1 Program-A issues a request to the DBMS (expressed in terms of sub-schema language);
- 2 DBMS looks at Program-A sub-schema, schema and physical description (these information are stored in tables);
- 3 DBMS determines which files must be accessed, which records are needed and how access is done;
- 4 DBMS issues instruction(s) (reads or writes) to the operating system;
- 5 Operating system causes data transfer between disk storage and main memory;
- 6 DBMS issues moves to transfer required fields;
- 7 DBMS returns control to Program-A (possibly with a completion code).

122

# Database Management System

## Overall Advantages

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

16

- ✓ Redundancy can be reduced.
- ✓ Inconsistencies can be avoided.
- ✓ Data can be shared.
- ✓ Standards can be enforced.
- ✓ Security restrictions can be applied.
- ✓ Integrity can be maintained.
- ✓ Conflicting requirements can be balanced.
- ✓ Improved performance due to speed of processing, reduction in paperwork, etc.
- ✓ Maintenance and retrieval of data are very easy.
- ✓ Is not solely dependent on the high level language (HLL) programming for use.
- ✓ Logical views of data stored can be easily created.

122



# Types of Databases

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

17

- 1 Relational Databases
- 2 Non Relational (NoSQL) Databases
- 3 Graph / Network Databases

## Based on other categorization:

- Columnar Databases
- Time-Series Databases
- Spatial Databases
- Hierarchical Databases
- Object-Oriented Databases
- Document Databases
- Key-Value Databases





# Types of Databases

## Relational Databases

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

### Introduction

### Introduction to DBMS

### Types of Databases

### Database Languages

### Basics of Relational Databases

### Hands-On Tutorial on DQL

### Database Design

### Entity Relationship Model

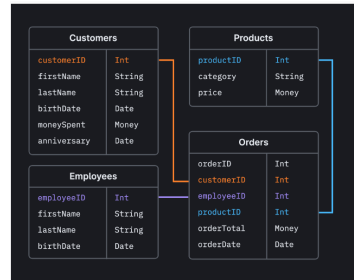
### Joins, Unions, and String Functions

### Normalization

18

- One way to store data related to each other in a pre-defined way.
- Organizes data into predefined structured tables
- At the time of the creation of the database, you can identify the relationships that exist between different entities or groups of data
- Rows represent records, and columns represent fields
- Where Relational Databases are used?
  - ▶ Websites-E-commerce (orders, products, customers)
  - ▶ Businesses, (transactions, accounts)
  - ▶ Mobile Applications
  - ▶ Any application requiring strict consistency and structured data
- Examples:
  - ▶ MySQL, PostgreSQL

Limitations: Not ideal for highly flexible or unstructured data



122

# Types of Databases

## Non Relational Databases

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and Scope

Introduction

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

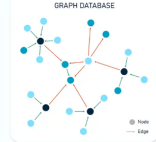
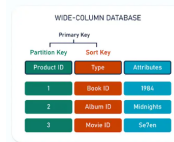
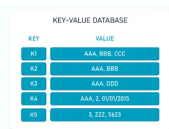
Entity Relationship Model

Joins, Unions, and String Functions

Normalization

19

- Does not use the traditional table-based relational database model
- Designed to be scalable and flexible
- Use variety of data models /schema
- Where Non Relational Databases are used?
  - Storing large amounts of data
  - Handling high traffic
  - Mobile Applications
  - Processing real-time data: Traffic data from large networks
- Examples:
  - Apache Cassandra, Hadoop, MongoDB



# Types of Databases

## Graph Databases

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

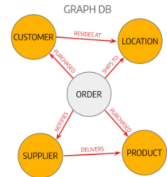
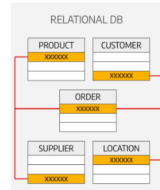
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

20

- Designed to represent and store data with complex relationships using structures called nodes (entities) and edges (relationships).
- A type of NoSQL database
- Optimized for connected data: Ideal for many-to-many relationships
- Flexible schema: Adapts easily to evolving structures
- Where Graph Databases are used?
  - ▶ Social Networks
  - ▶ Fraud Detection
  - ▶ Supply Chain Logistics
- Examples:
  - ▶ Amazon Neptune: managed graph DB service on AWS
  - ▶ TigerGraph – Enterprise-grade graph analytics platform



# Types of Databases

## Graph Databases

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

### Introduction

### Introduction to DBMS

### Types of Databases

### Database Languages

### Basics of Relational Databases

### Hands-On Tutorial on DQL

### Database Design

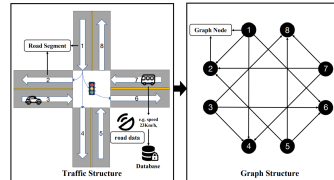
### Entity Relationship Model

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# Types of Databases

## NoSQL - Columnar Databases

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

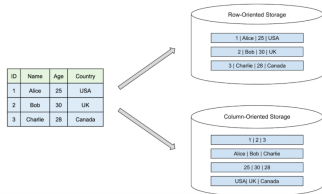
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

21

- Designed to store data by columns rather than rows
- Enables fast analytical queries over massive datasets.
- Well-suited for high-performance analytical workloads.
- Use-cases:
  - ▶ Business intelligence dashboards
  - ▶ Reporting and KPI tracking
  - ▶ Supply Chain Logistics
- Examples:
  - ▶ Apache HBase – Wide-column NoSQL database built on top of Hadoop
  - ▶ Amazon Redshift – Managed data warehouse with columnar storage
  - ▶ Google BigQuery – Serverless, highly scalable cloud data warehouse



122

# Examples of Databases

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

22

- Oracle: Widely used relational database management system
- MySQL: Open-source relational database management system
- Microsoft SQL Server: Relational database management system by Microsoft
- PostgreSQL Open-source relational database management system
- SQLite Lightweight, embedded relational database system
- MongoDB: NoSQL database for handling unstructured data
- Cassandra: Distributed NoSQL database for scalability
- Redis: In-memory data structure store and cache
- Amazon Aurora: Cloud-native relational database service
- Neo4j: Graph database for managing interconnected data
- Apache HBase: Wide-column NoSQL database built on top of Hadoop
- Amazon Redshift: Columnar storage
- Amazon Neptune: managed graph DB service on AWS

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

23

- Understanding the types of databases is crucial for designing scalable, efficient, and reliable applications.
- Each type is optimized for a specific use case.
- The key is choosing the right database type for your work.
- Example:
  - ▶ A transactional financial system → go with a relational database
  - ▶ A social network or fraud detection system → leverage a graph database
  - ▶ A time-based monitoring tool → implement a time-series database
  - ▶ An analytics dashboard → opt for a columnar database

122

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Specialized sets of commands and instructions used to define, manipulate and control data within a database.

- 1 Data Definition Language (DDL)
- 2 Data Manipulation Language (DML)
- 3 Data Control Language (DCL)
- 4 Transaction Control Language (TCL)
- 5 Data Query Language (DQL)

24

122



# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

**Database Languages**

Data Definition Language (DDL)  
Data Manipulation Language (DML)  
Data Query Language (DQL)  
Data Control Language (DCL)  
Transaction Control Language (TCL)

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

25

122

## Data Definition Language (DDL):

- Define how the data should reside in the database.
- **CREATE:** to create a database and its objects like (table, index, views, store procedure, function and triggers)
- **ALTER:** alters the structure of the existing database
- **DROP:** delete objects from the database
- **TRUNCATE :** remove all records from a table, including all spaces allocated for the records are removed
- **COMMENT :** add comments to the data dictionary
- **RENAME :** rename an object

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

**Database Languages**

Data Definition Language (DDL)  
Data Manipulation Language (DML)  
Data Query Language (DQL)  
Data Control Language (DCL)  
Transaction Control Language (TCL)

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

27

122

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Data Manipulation Language (DML):

- Used to manipulate the data stored in database
- INSERT: insert data into a table
- UPDATE: updates existing data within a table
- DELETE: delete objects from the database
- MERGE : Merge the records from a table

28

122

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

**Database Languages**

Data Definition Language (DDL)  
Data Manipulation Language (DML)  
Data Query Language (DQL)  
Data Control Language (DCL)  
Transaction Control Language (TCL)

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

29

122

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Data Query Language (DQL):

- Used to retrieve data from a database without modifying it.
- **SELECT:** select a particular column or table from the database

30

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

**Database Languages**

Data Definition Language (DDL)  
Data Manipulation Language (DML)  
Data Query Language (DQL)  
Data Control Language (DCL)  
Transaction Control Language (TCL)

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

31

122

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

32

## Data Control Language (DCL):

- Used to manage access permissions, ensuring data security.
- **GRANT:** Provides specific privileges to a user (e.g., SELECT, INSERT).
- **REVOKE:** Removes previously granted permissions from a user.

122



# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

**Database Languages**

Data Definition Language (DDL)

Data Manipulation Language (DML)

Data Query Language (DQL)

Data Control Language (DCL)

Transaction Control Language (TCL)

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

33

122

## Transaction Control Language (TCL):

- Used to oversee transactional data to maintain consistency, reliability and
- COMMIT: Saves all changes made during a transaction.
- ROLLBACK: Undoes changes made during a transaction.
- SAVEPOINT: Sets a point within a transaction to which one can later roll back.

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

**Basics of  
Relational  
Databases**

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

**Basics of Relational Databases**

Focus of the Course

Elements of Database

Domain/ variable Types in SQL

Database Schema

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

35

122

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

**Basics of  
Relational  
Databases**

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

36

122

- The course focuses on Relational Database System
  - ▶ Data Definition Language (DDL)
  - ▶ Data Manipulation Language (DML)
  - ▶ Data Query Language (DQL)

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

**Basics of  
Relational  
Databases**

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

**Basics of Relational Databases**

Focus of the Course

Elements of Database

Domain/ variable Types in SQL

Database Schema

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

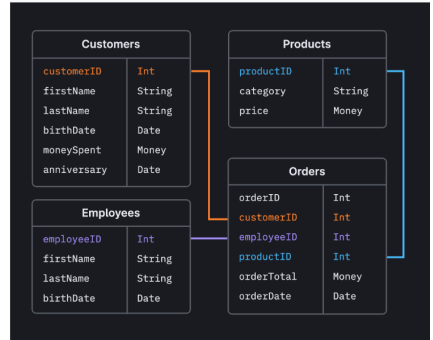
37

122

# Relational Database System

## Elements of Database / Basic Terms

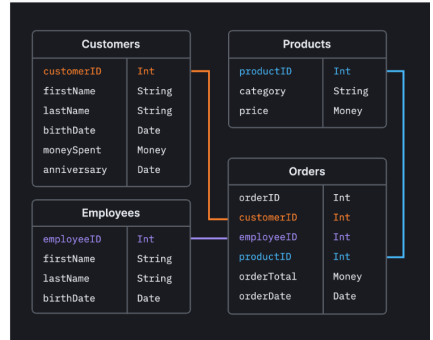
- 1 Tables
- 2 Columns / Attribute / Variable
  - ▶ Degree: Number of columns
- 3 Tuple/ Row / Record
  - ▶ Cardinality: Number of rows
- 4 Domain: Legal values of attributes
- 5 Entity: Object in the real world
- 6 Relationship: An association among several entities
- 7 Key
  - ▶ Primary Key: Unique identifier of each record. eg: `product_ID`
  - ▶ Foreign Key: A column that refers to a primary key in another table. `employee_ID`



# Relational Database System

## Elements of Database / Basic Terms

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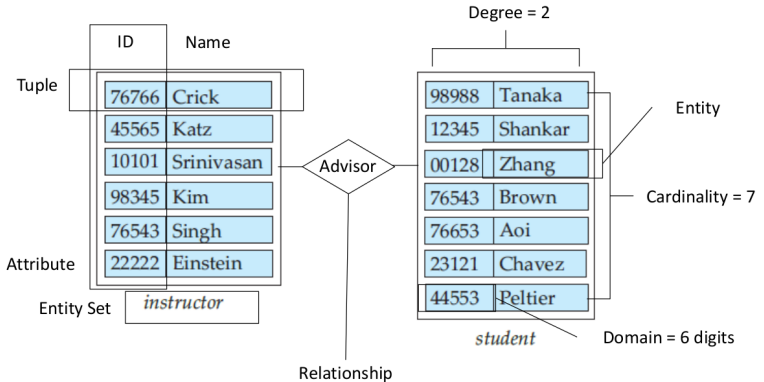


## Database Schema

- ▶ The schema is the structure that we define for our data. The schema defines the tables, relationships between tables, fields, and indexes

# Relational Database System

Elements of Database / Basic Terms





# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

**Basics of  
Relational  
Databases**

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

**Basics of Relational Databases**

Focus of the Course

Elements of Database

Domain/ variable Types in SQL

Database Schema

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

40

122

## Variable Types

- `char(n)`: Fixed length character string, with user-specified length  $n$ .
- `varchar(n)`: Variable length character strings, with user-specified maximum length  $n$ .
- `int.`: Integer (a finite subset of the integers that is machine-dependent).
- `numeric(p,d)`: Fixed point number, with user-specified precision of  $p$  digits, with  $d$  digits to the right of decimal point.
  - ex., `numeric(3,1)`, allows 44.5 to be stored exactly, but not 444.5 or 0.3
- `float(n)`: Floating point number, with user-specified precision of at least  $n$  digits.
- `real`, `double precision`: Floating point and double-precision floating point numbers, with machine-dependent precision.

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

**Basics of  
Relational  
Databases**

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

**Basics of Relational Databases**

Focus of the Course

Elements of Database

Domain/ variable Types in SQL

Database Schema

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

42

122

# Database Schema

## Database Systems and Information Integration

Dr. Rushikesh Amrutsamanvar

Objective and Scope

Introduction

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

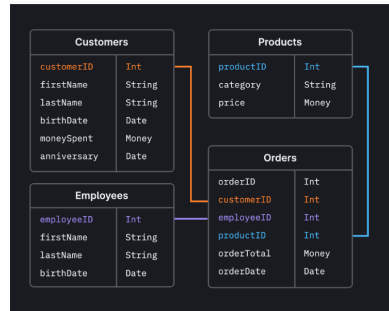
Entity Relationship Model

Joins, Unions, and String Functions

Normalization

- Defines the structure and organization of data within a database.
- Blueprint for how data is:
  - ▶ Logically organized
  - ▶ Physically Stored
  - ▶ Accessed
  - ▶ Manipulated

ensuring consistency and integrity throughout the system.



# Database Schema

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

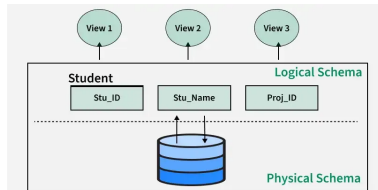
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- Blueprint for how data is:
  - ▶ Logically organized
  - ▶ Physically Stored
  - ▶ Accessed
  - ▶ Manipulated

ensuring consistency and integrity throughout the system.

### Types of Database Schema:

- 1 Physical Database Schema: defines how data is stored in the storage system
- 2 Logical Database Schema: defines the logical structure of the data
- 3 View Database Schema: defines users interact with the database



# Now lets start learning by doing !!!

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial 44  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Installation of MySQL workbench
- We will start with Querying the data :: DQL

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial on DQL 45

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

**Hands-On Tutorial on DQL**  
**Clauses in SQL**  
**Mathematical Operators**

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

# How SQL Thinks

## The Data Pipeline

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

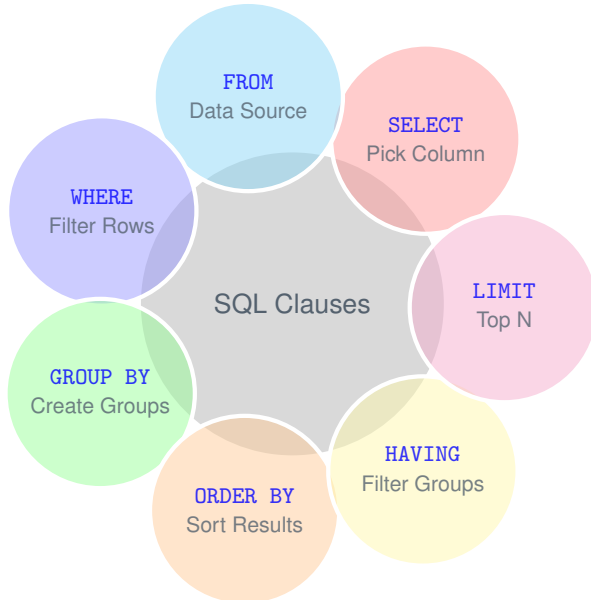
Hands-On Tutorial 46  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization





# SQL Query Processing Flow

Logical Execution Order: SQL is written top-down, but executed bottom-up logically

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

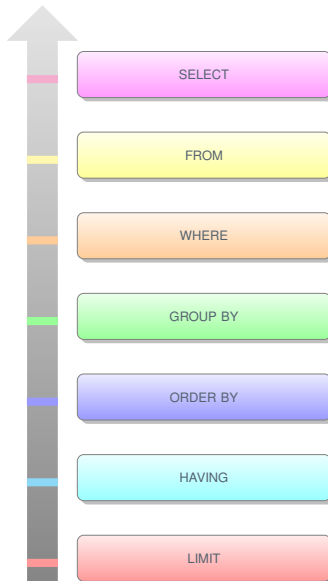
Hands-On Tutorial **47**  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization



# Clauses

SQL is written top-down, but executed bottom-up logically.

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

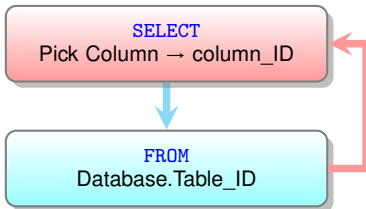
Hands-On Tutorial 48  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization



# Clauses

SQL is written top-down, but executed bottom-up logically.

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

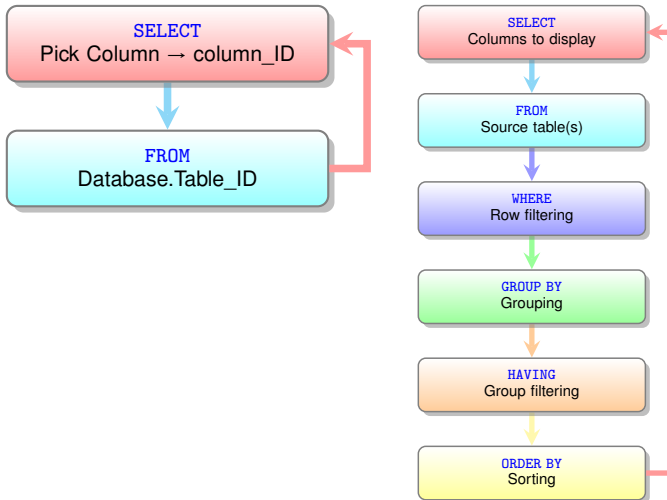
Hands-On Tutorial 48  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization



# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial on DQL 49

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

**Hands-On Tutorial on DQL**  
**Clauses in SQL**  
**Mathematical Operators**

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

# Mathematical Operators and Operations

## Addition and Subtraction

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

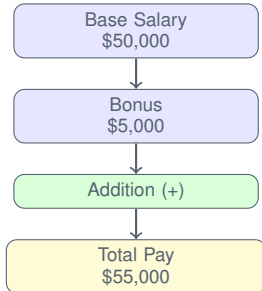
Hands-On Tutorial  
on DQL

Database Design

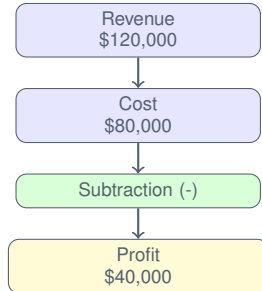
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization



**SELECT** salary + bonus **FROM**  
employees;



**SELECT** revenue - cost **FROM** sales;

# Mathematical Operators

## Multiplication and Division

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

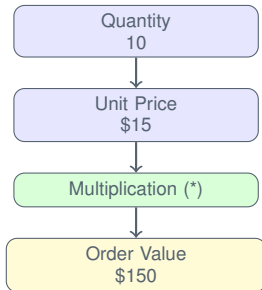
Entity Relationship  
Model

Joins, Unions, and  
String Functions

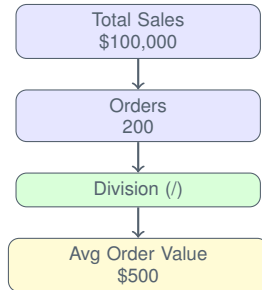
Normalization

51

122



```
SELECT quantity * price FROM orders;
```



```
SELECT total_sales / orders FROM summary;
```

# Mathematical Operators

## Modulus and Order of Operations

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

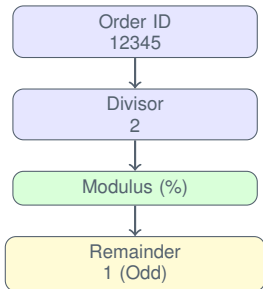
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

52

122



```
SELECT order_id % 2 FROM orders;
```



```
SELECT (price - discount) * tax  
FROM sales;
```

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 53

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

## Database Design

Fundamentals

What is good database design?

Database Design Lifecycle

Entity Relationship Model

Joins, Unions, and String Functions

Normalization 122



# Database Design

## What is it?

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 54

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- A set of procedures or collection of tasks taken to build the structure of database.
- A properly structured database guarantees efficiency, data accuracy, and usability.
- Essential for anyone who is involved into designing, managing, or optimizing databases.

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 55

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

## Database Design

Fundamentals

What is good database design?

Database Design Lifecycle

Entity Relationship Model

Joins, Unions, and String Functions

Normalization 122

# Database Design

## What is good database design?

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

56

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Critical aspects of good database design:

- Data should be **consistent** and its **integrity** must be maintained
  - ▶ Establish procedures and enforce to minimize errors
  - ▶ Ensure the accuracy of data
  - ▶ Integrity  $\equiv$  Quality, Consistency  $\equiv$  Correctness
- **Low redundancy**
  - ▶ Data Normalization
    - No duplicates
    - Minimal information repetition
    - Faster searching through indices
  - ▶ Divide information into subject-based tables
- **Security** must be enhanced
  - ▶ Protect sensitive data from unauthorized access.
  - ▶ Access controls, authentication mechanisms, and encryption techniques to safeguard data at rest and in transit.

# Database Design

## What is good database design?

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 57

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Data should be stored in fragmented bits of information in the most atomic format possible.
  - ▶ Breaking it into the smallest possible indivisible units
  - ▶ Atomicity  $\equiv$  each cell holds a single, indivisible value (e.g., one name, not a comma-separated list).
  - ▶ Fragmentation  $\equiv$  Splitting large data into smaller pieces
- Streamlining the database
  - ▶ Transactions should be executed in a way that maintains consistency, correctness, and reliability even in case of failures / errors.
  - ▶ Maintain ACID / BASE Properties of database during transactions
    - ACID  $\rightarrow$  Atomicity, Consistency, Isolation, and Durability.
    - BASE  $\rightarrow$  more used in case of NoSQL databases
    - Basically Available, Soft State, Eventually Consistent

# SQL Clauses

## FROM and SELECT

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

58

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

122

sales table  
(order\_id, product, revenue)



FROM sales



All Sales Data

**FROM = where data comes from**

All Columns



SELECT product, revenue



Chosen Columns

**SELECT = column chooser**

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 59

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

## Database Design

Fundamentals

What is good database design?

Database Design Lifecycle

Entity Relationship Model

Joins, Unions, and String Functions

Normalization 122

# Database Design Lifecycle

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

1

## Requirement Analysis:

1. Understand the purpose of the database
2. Organize the required information

2

## Logical Design:

1. Coming up with a high-level design of database based on requirements
2. Data Models: to get a high-level overview of database.
3. Define the Entity, Attributes and Relationships.
4. Normalization



# Database Design Lifecycle

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 60

Entity Relationship  
Model

Joins, Unions, and  
String Functions

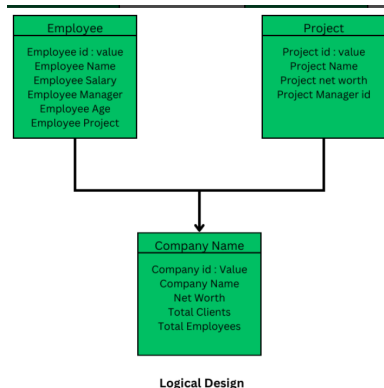
Normalization

## 1 Requirement Analysis:

1. Understand the purpose of the database
2. Organize the required information

## 2 Logical Design:

1. Coming up with a high-level design of database based on requirements
2. Data Models: to get a high-level overview of database.
3. Define the Entity, Attributes and Relationships.
4. Normalization





# Database Design Lifecycle

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 61

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## 3 Physical Design:

- ▶ Implementation of logical design.
  - ▶ Demonstrate the database structure along with all the columns & their data types, rows, relations, relationships among data
1. Entities are converted into tables or relations that consist of their properties (attributes)
  2. Apply integrity constraints: establish foreign key, unique key, and composite key relationships among the data. And apply various constraints.
  3. Apply normalization.

## 4 Implementation

## 5 Test



# Database Design Lifecycle

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design 61

Entity Relationship  
Model

Joins, Unions, and  
String Functions

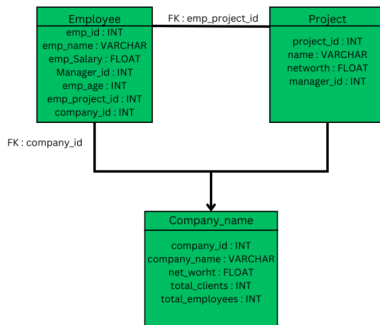
Normalization

## 3 Physical Design:

- ▶ Implementation of logical design.
  - ▶ Demonstrate the database structure along with all the columns & their data types, rows, relations, relationships among data
1. Entities are converted into tables or relations that consist of their properties (attributes)
  2. Apply integrity constraints: establish foreign key, unique key, and composite key relationships among the data. And apply various constraints.
  3. Apply normalization.

## 4 Implementation

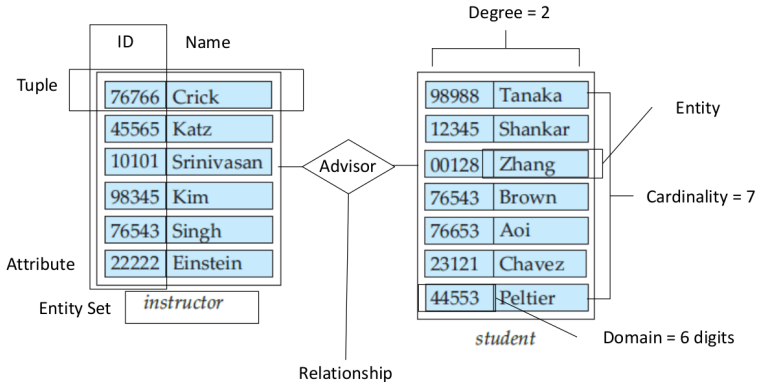
## 5 Test



Physical Design

# Database Design

revisiting basic taxonomy used in database design



- Schema: a logical container that defines the structure
- Data Models: provide us with visual modeling techniques to visualize the data & the relationship that exists among those data.
  - ▶ Example: ER Model, Network Model, Object Oriented Model, Hierarchical model, etc.

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Basics

Components of ER Diagram

Relationships, Degree and Cardinality

Building ERD

Practice Example

Joins, Unions, and String Functions

# Entity Relationship (ER) Model

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- The Entity-Relationship Model (ER Model) is a conceptual model for designing a databases.
- Chen Pin-Shan proposed this method in 1976
- It represents the logical structure of a database.
  - ▶ **Entity:** Objects that is stored as data such as Student, Course or Company.
  - ▶ **Attribute:** Properties that describes an entity such as StudentID, CourseName, or Employee Email.
  - ▶ **Relationship:** A connection between entities such as "a Student enrolls in a Course".
- The graphical representation of this model is called an Entity-Relation Diagram (ERD).

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Basics

Components of ER Diagram

Relationships, Degree and Cardinality

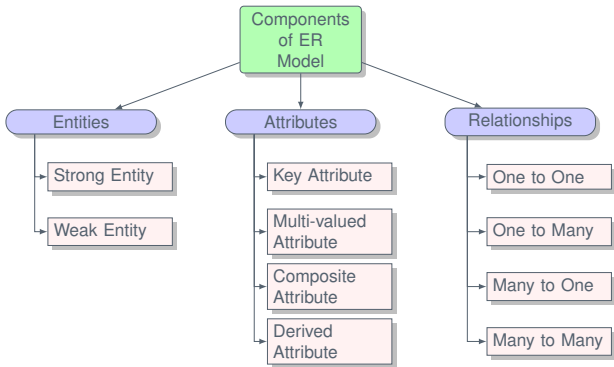
Building ERD

Practice Example

Joins, Unions, and String Functions

# Entity Relationship (ER) Model

## Components



- **Strong Entity:** Entity does not depend on any other Entity in the Schema for its identification.
  - ▶ It has a **primary key** that ensures its uniqueness
  - ▶ Represented by rectangle in ER Diagram
- **Weak Entity:** Entity that cannot be uniquely identified by its own attributes alone.
  - ▶ Represented by a double rectangle

# Entity Relationship (ER) Model

## Components

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

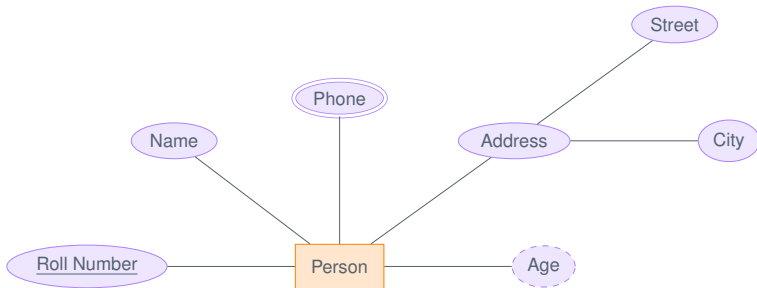
Hands-On Tutorial  
on DQL

Database Design

Entity Relationship Model

Joins, Unions, and  
String Functions

Normalization



- **Key Attribute:** The attribute which uniquely identifies each entity in the entity set .
  - ▶ Represented by an oval with an underline.
  - ▶ In this example: Roll Number

- **Composite Attribute:** An attribute composed of many other attributes
  - ▶ Example: Address - it encompasses Street, City, State, and Country
  - ▶ Represented by an oval comprising of ovals.



# Entity Relationship (ER) Model

## Components

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

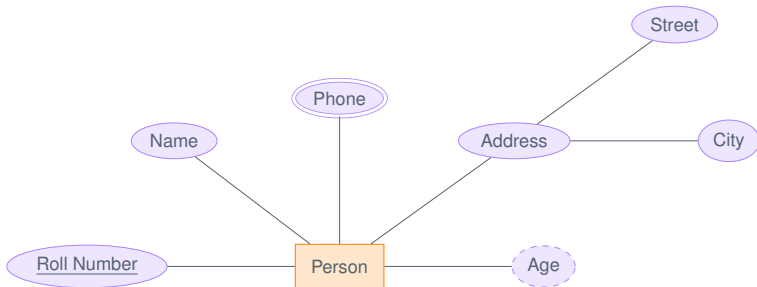
Hands-On Tutorial  
on DQL

Database Design

Entity Relationship Model

Joins, Unions, and  
String Functions

Normalization



- **Multivalued Attribute:** An attribute consisting of more than one value for a given entity.
  - Represented by a double oval
  - In this example: Phone Number

- **Derived Attribute:** An attribute that can be derived from other attributes of the entity type is known as a derived attribute..
  - Represented by a dashed oval
  - In this example: Age

# ER Diagram Notations

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Figures          | Symbols                                                                           | Represents                                                              |
|------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Rectangle        |  | Entities in ER Model                                                    |
| Ellipse          |  | Attributes in ER Model                                                  |
| Diamond          |  | Relationships among Entities                                            |
| Line             |  | Attributes to Entities and Entity Sets with<br>Other Relationship Types |
| Double Ellipse   |  | Multi-Valued Attributes                                                 |
| Double Rectangle |  | Weak Entity                                                             |

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Basics

Components of ER Diagram

Relationships, Degree and Cardinality

Building ERD

Practice Example

Joins, Unions, and String Functions

# Relationship Type and Relationship Set

## Degree of a Relationship Set

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- A Relationship Type represents the association between entity types.
- Type:
  - 1 **Unary/Recursive Relationship:** only ONE entity set participating in a relation
  - 2 **Binary Relationship:** when there are TWO entities set participating in a relationship
  - 3 **Ternary Relationship:** when there are three entity sets participating in a relationship
  - 4 **N-ary Relationship:** when there are n entities set participating in a relationship

# Relationship Type and Relationship Set

## Mapping Cardinalities

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

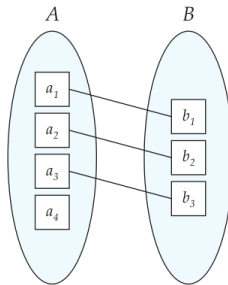
Database Design

Entity Relationship  
Model

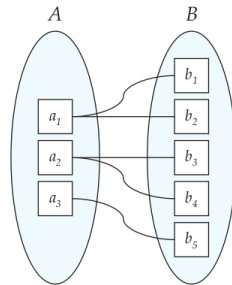
Joins, Unions, and  
String Functions

Normalization

The maximum number of times an entity of an entity set participates in a relationship set is known as cardinality.



- **One-to-one:** An entity in  $A$  is associated with at most one entity in  $B$ , and an entity in  $B$  is associated with at most one entity in  $A$ .



- **One-to-many:** An entity in  $A$  is associated with any number (zero or more) of entities in  $B$ . An entity in  $B$ , however, can be associated with at most one entity in  $A$ .

# Relationship Type and Relationship Set

## Mapping Cardinalities

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

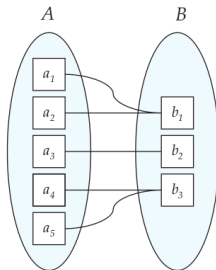
Hands-On Tutorial  
on DQL

Database Design

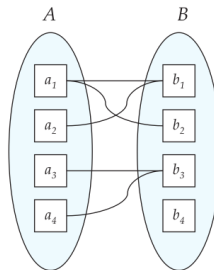
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization



- **Many-to-one:** An entity in  $A$  is associated with at most one entity in  $B$ . An entity in  $B$ , however, can be associated with any number (zero or more) of entities in  $A$ .



- **Many-to-many:** An entity in  $A$  is associated with any number (zero or more) of entities in  $B$ , and an entity in  $B$  is associated with any number (zero or more) of entities in  $A$ .

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Basics

Components of ER Diagram

Relationships, Degree and Cardinality

Building ERD

Practice Example

Joins, Unions, and String Functions

# How to build an ER Diagram

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- 1 Identify Entities:** The very first step is to identify all the Entities. Represent these entities in a Rectangle and label them accordingly.
- 2 Identify Relationships:** The next step is to identify the relationship between them and represent them accordingly using the Diamond shape. Ensure that relationships are not directly connected to each other.
- 3 Add Attributes:** Attach attributes to the entities by using ovals. Each entity can have multiple attributes (such as name, age, etc.), which are connected to the respective entity.
- 4 Define Primary Keys:** Assign primary keys to each entity. These are unique identifiers that help distinguish each instance of the entity. Represent them with underlined attributes.
- 5 Remove Redundancies:** Review the diagram and eliminate unnecessary or repetitive entities and relationships.
- 6 Review for Clarity:** Review the diagram make sure it is clear and effectively conveys the relationships between the entities.



# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Basics

Components of ER Diagram

Relationships, Degree and Cardinality

Building ERD

Practice Example

Joins, Unions, and String Functions

# Practice Example

Designing the database is like Art, its unique!

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Lets see the raw data we have.
- Open the sheet raw data that is shared with you.
- What are the errors?

# Practice Example

## How is this dataset

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- 1 Missing employee\_id (row 13).
- 2 Duplicate employee\_id for Tom (row 3 and 14).
- 3 Inconsistent salary formats: 50,000, fifty k, 60000, 55,000 (mix of commas, text).
- 4 department\_name and dept\_id inconsistent (one, 01, NULL, 2,3, Animal Control;Public Works).
- 5 Multi-valued fields: phone\_numbers has several numbers separated by semicolons. job\_title sometimes contains multiple jobs separated by semicolon.
- 6 Inconsistent emails and bad formats (jerry[at]example.com).
- 7 Manager stored by name (text) rather than as a relationship to an employee id.
- 8 Some fields are NULL or empty; some dates missing.
- 9 Status contains contract, active — possible domain but not normalized.

# What are the components of this database?

## Entity

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

**Entity Relationship  
Model**

Joins, Unions, and  
String Functions

Normalization

1 What are the Entities?:: real world objects/ concept that we want to store information about

79

# What are the components of this database?

## Entity

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

1 What are the Entities?:: real world objects/ concept that we want to store information about

- Entities usually correspond to nouns in requirements (employee, department, project, invoice).

# What are the components of this database?

## Entity

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

1 What are the Entities?:: real world objects/ concept that we want to store information about

- Entities usually correspond to nouns in requirements (employee, department, project, invoice).

2 What are strong entities?

79

122

# What are the components of this database?

## Entity

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- 1 What are the Entities?: real world objects/ concept that we want to store information about
  - ▶ Entities usually correspond to nouns in requirements (employee, department, project, invoice).
- 2 What are strong entities?
  - ▶ **Employee** is a strong entity — it can be uniquely identified by employee\_id
  - ▶ **Department** is also a strong entity — identified by department\_id.
- 3 What are weak entities and why?

# What are the components of this database?

## Attributes

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

1 What are the Attributes?:: a property or characteristic of an entity.

- Identify different types of Attributes from this database

80

122



# What are the components of this database?

## Attributes

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

### Introduction

### Introduction to DBMS

### Types of Databases

### Database Languages

### Basics of Relational Databases

### Hands-On Tutorial on DQL

### Database Design

### Entity Relationship Model

### Joins, Unions, and String Functions

### Normalization

1 What are the Attributes?: a property or characteristic of an entity.

- ▶ Identify different types of Attributes from this database

2 Types of attributes:

- ▶ Simple / Atomic: cannot be subdivided further. E.g., `first_name`, `salary` (if stored as a single number).
- ▶ Composite: made of smaller parts. E.g., new variable `name` could be `first_name + last_name`.
- ▶ Multi-valued: can have multiple values for the same entity. E.g., `phone_numbers` (an employee can have several numbers).
- ▶ Derived: computed from other attributes. E.g., `age` derived from `birth_date`.
- ▶ Nullability: attributes may allow NULL (missing) values — we should minimize NULLs for required fields.

3 In our raw table:

# What are the components of this database?

## Attributes

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

### Objective and Scope

### Introduction

### Introduction to DBMS

### Types of Databases

### Database Languages

### Basics of Relational Databases

### Hands-On Tutorial on DQL

### Database Design

### Entity Relationship Model

### Joins, Unions, and String Functions

### Normalization

1 What are the Attributes?: a property or characteristic of an entity.

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- ▶ Derived: computed from other attributes. E.g., `age` derived from `birth_date`.
- ▶ Nullability: attributes may allow NULL (missing) values — we should minimize NULLs for required fields.

3 In our raw table:

- ▶ `phone_numbers` is multi-valued (bad in a single cell).
- ▶ `age` and `birth_date` are redundant: `age` is derived and will become stale.
- ▶ `textttname` can be made composite (full name)

# What are the components of this database?

## Attributes

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Column          | Type of Attribute                             | Notes                                   |
|-----------------|-----------------------------------------------|-----------------------------------------|
| employee_id     | Identifier / Key                              | numeric or NULL (error)                 |
| name            | Categorical (nominal)                         | text                                    |
| age             | Numerical (discrete)                          | sometimes missing                       |
| gender          | Categorical (nominal)                         | Male/Female/Other                       |
| birth_date      | Date                                          | missing for some                        |
| job_title       | Categorical (nominal)                         | even has multiple roles in row 15       |
| salary          | Numerical but dirty                           | 50,000, fifty k → inconsistent          |
| dept_id         | Categorical / should be FK                    | Wrong formats: NULL, "one", "2,3", "01" |
| department_name | Categorical                                   | Duplicates/inconsistencies              |
| phone_numbers   | Multi-valued attribute                        | Should be a separate weak entity        |
| email           | Categorical                                   | inconsistent formats                    |
| manager_name    | Categorical / Should be reference to employee | Broken foreign key                      |
| start_date      | Date                                          | looks clean                             |
| status          | Categorical                                   | active/contract                         |

# ER in the Parks and Recreation Dataset

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Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship Model

Joins, Unions, and  
String Functions

Normalization

Our raw employee dataset contains mixed information. From the data, we can observe multiple types of relationships:

## 1. Employee ↔ Department

- Normally, each employee belongs to one department (1:N relationship)
- Example: Leslie Knope → Parks and Recreation
- Exceptions:
  - ▶ Row 15 (Multi Dept Person) → Employee belongs to multiple departments
  - ▶ This makes it a Many-to-Many relationship
- **Solution:** Create a weak entity `employee_department` for M:N mapping

## 2. Employee ↔ Phone Numbers

- Employees can have multiple phone numbers
- Example: Leslie Knope → 555-0101, 555-0102
- **Type:** One-to-Many
- **Solution:** Create a weak entity `employee_phone_numbers`

### 3. Employee ↔ Manager (Recursive Relationship)

- Manager is also an employee
- Each employee may have zero or one manager
- One manager can supervise many employees (1:N)
- Example:
  - ▶ Leslie Knope manages Tom Haverford, April Ludgate, Craig Middlebrooks
  - ▶ Chris Traeger manages Ann Perkins, Ben Wyatt, Mark Brendanawicz

### 4. Notes on Raw Data Issues

- Many inconsistencies in department IDs, manager names, and phone numbers
- Need to normalize the data

# Deciding Which Tables to Create

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Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Rule of Thumb

- Each **strong entity** becomes a table.
- **Multi-valued attributes** become separate tables (child tables).
- **Composite attributes** are split into atomic columns.
- **Relationships with attributes** or **M:N cardinality** often become tables.
- **Derived attributes** should be removed or computed on the fly.

## From the Raw Dataset, Tables Should Include

- Employee – personal details
- Department – lookup
- Type\_of\_Job – occupation, salary, time-based info
- Phone\_Number – multi-valued attribute
- Optionally Employment\_Status – enumerated domain (active, contract, terminated)

# Summary of Relationships

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Relationship             | Type                  | Implementation                          |
|--------------------------|-----------------------|-----------------------------------------|
| Employee → Department    | Many-to-Many          | employee_department table (weak entity) |
| Employee → Phone Numbers | One-to-Many           | employee_phones table (weak entity)     |
| Employee → Manager       | Recursive One-to-Many | manager_id FK referencing employee_id   |

## Takeaway

Understanding relationships in messy raw data is crucial for:

- Normalization
- Designing proper keys and foreign keys
- Avoiding duplicate and inconsistent data

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Normalize the dataset into 1NF, 2NF, 3NF
- Create separate tables for multi-valued and composite attributes
- Establish proper primary and foreign keys



# SQL Joins

## Column Operations

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

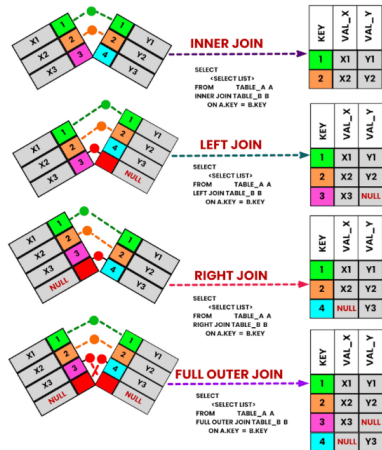
Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- **Inner Join:** When you need rows that have matching values in both tables. This is the most performance-efficient join when you only care about matching rows.
- **Left Outer Join:** When you want all records from the left table to show up even if there is no corresponding entry in the right table.
- **Right Outer Join:** Similar to the left. But when you care about returning all the entries in the right table even if they have no matching entries in the left.
- **Full Outer Join:** When you need all records from both tables.



# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Joins

Unions

String Functions

Normalization

88

122

# Joins

## INNER JOIN

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

**Table A**

|   |
|---|
| 1 |
| 2 |
| 3 |

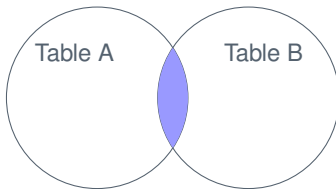
**Table B**

|   |
|---|
| 1 |
| 2 |
| 4 |

**Result**

|             |
|-------------|
| 1   X1   Y1 |
| 2   X2   Y2 |

**INNER JOIN:** Only rows with matching keys appear in the result.



### SQL Query

```
SELECT A.id, A.value, B.value
FROM Table_A A
INNER JOIN Table_B B
ON A.id = B.id;
```

**INNER JOIN** is the default JOIN in SQL

# Joins

## LEFT JOIN

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

90

122

Table A

|   |
|---|
| 1 |
| 2 |
| 3 |

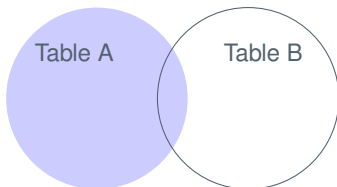
Table B

|   |
|---|
| 1 |
| 2 |
| 4 |

Result

|               |
|---------------|
| 1   X1   Y1   |
| 2   X2   Y2   |
| 3   X3   NULL |

**LEFT JOIN:** All rows from Table A, unmatched rows get NULLs.



### SQL Query

```
SELECT A.id, A.value, B.value
FROM Table_A A
LEFT JOIN Table_B B
ON A.id = B.id;
```

The first selected Table is LEFT  
Table

# Joins

## OUTER JOIN

Database Systems  
and Information  
Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Table A

|   |
|---|
| 1 |
| 2 |
| 3 |

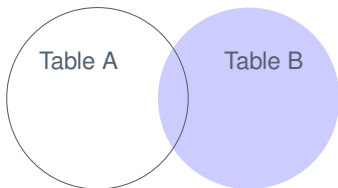
Table B

|   |
|---|
| 1 |
| 2 |
| 4 |

Result

|               |
|---------------|
| 1   X1   Y1   |
| 2   X2   Y2   |
| 4   NULL   Y3 |

**RIGHT JOIN:** All rows from Table B, unmatched rows get NULLs



### SQL Query

```
SELECT A.id, A.value, B.value
FROM Table_A A
RIGHT JOIN Table_B B
ON A.id = B.id;
```

The first selected Table is RIGHT  
Table

# Joins

## RIGHT JOIN

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Table A

|   |
|---|
| 1 |
| 2 |
| 3 |

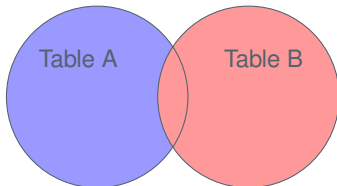
Table B

|   |
|---|
| 1 |
| 2 |
| 4 |

Result

|               |
|---------------|
| 1   X1   Y1   |
| 2   X2   Y2   |
| 3   X3   NULL |
| 4   NULL   Y3 |

**OUTER JOIN:** All rows from Table A & B, unmatched rows get NULLs



### SQL Query

```
SELECT A.id, A.value, B.value
FROM Table_A A
OUTER JOIN Table_B B
ON A.id = B.id;
```

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Joins  
Unions  
String Functions

Normalization

93

122

# Unions

## Row Operations

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

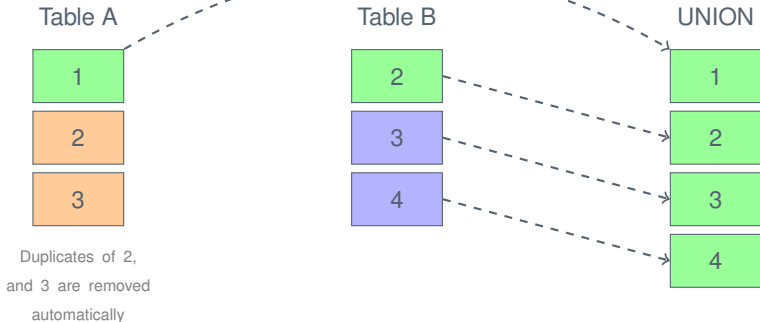
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

94

122



**UNION:** Combines rows from both tables and removes duplicates.

```
SELECT A.id, A.value, B.value
FROM Table_A A
UNION
SELECT A.id, A.value, B.value
FROM Table_B B;
```



# Unions

## Row Operations

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

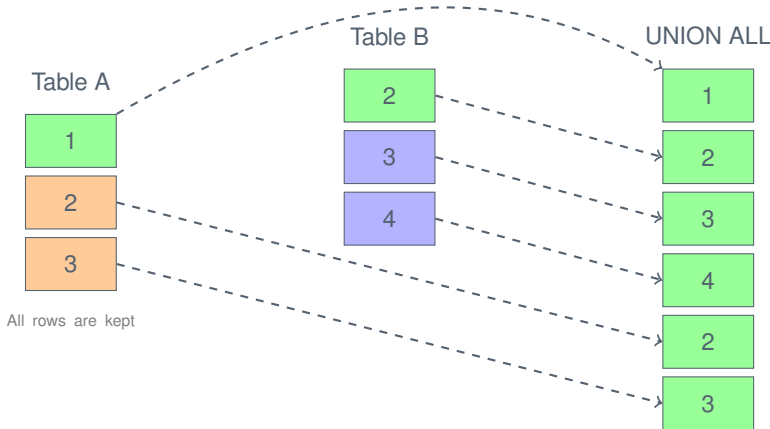
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

95

122



**UNION:** Combines rows from both tables and keeps the duplicates.

```
SELECT A.id, A.value, B.value FROM Table_A A
UNION ALL
SELECT A.id, A.value, B.value FROM Table_B B;
```

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Joins  
Unions  
String Functions

Normalization

96

122

# String Functions

## UPPER() and LOWER()

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

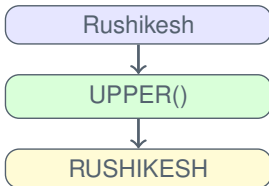
Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

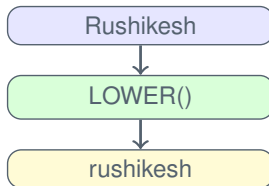
Joins, Unions, and  
String Functions

Normalization



```
SELECT UPPER('Rushikesh');
```

**Result:** RUSHIKESH



```
SELECT LOWER('Rushikesh');
```

**Result:** rushikesh

# String Functions

## TRIM() and REPLACE()

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

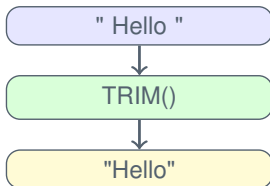
Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

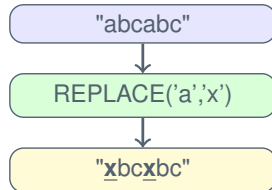
Joins, Unions, and  
String Functions

Normalization



```
SELECT TRIM(' Hello ');
```

**Result:** Hello



```
SELECT REPLACE('abcabc', 'a', 'x');
```

**Result:** xbcxabc

# String Functions

## CONCAT() and SUBSTRING()

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

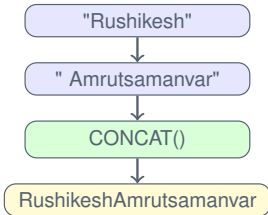
Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

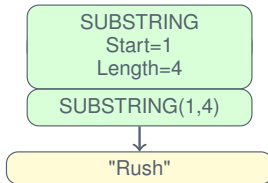
Joins, Unions, and  
String Functions

Normalization



```
SELECT CONCAT('Rushikesh',  
Amrutsamanvar');
```

**Result:** RushikeshAmrutsamanvar



```
SELECT SUBSTRING('Rushikesh',1,4);
```

**Result:** Rush

# SQL Query Flow

## FROM, SELECT, WHERE

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

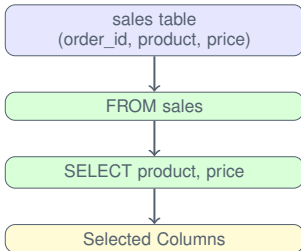
Hands-On Tutorial  
on DQL

Database Design

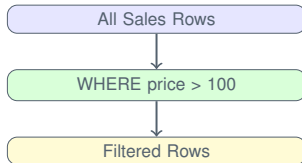
Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization



**SELECT** product, price\FROM sales;



**WHERE** price > 100;

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

Motivation

Overview of Normal Forms

Real-Word Example

101

122

# Normalization

## Why Do We Need Normalization?

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Real-world data is often stored in an unstructured way
- This leads to:
  - ▶ Data redundancy
  - ▶ Inconsistencies
  - ▶ Update anomalies
  - ▶ Insert anomalies
  - ▶ Delete anomalies
- Normalization is a systematic method to avoid these problems



# What Is Normalization?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Definition

Normalization is the process of organizing data in a relational database by decomposing tables according to well-defined rules (normal forms).

- Goal: logical data consistency
- Reduce redundancy
- Improve maintainability
- Preserve data integrity

# Normalization and Functional Dependencies

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Normalization is based on **functional dependencies**

- A functional dependency:

$$A \rightarrow B$$

means:

*Attribute A uniquely determines attribute B*

- Normal forms define which dependencies are allowed

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

Motivation

Overview of Normal Forms

Real-Word Example

105

122

# Overview of Normal Forms

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Normal Form | Main Goal                          |
|-------------|------------------------------------|
| 1NF         | Atomic values, no repeating groups |
| 2NF         | No partial dependencies            |
| 3NF         | No transitive dependencies         |

| Normal Form | What You Fix                   | Main Question to Ask                     |
|-------------|--------------------------------|------------------------------------------|
| 1NF         | Repeating groups, multi-values | "Is every value atomic?"                 |
| 2NF         | Partial dependencies           | "Does every attribute depend whole key?" |
| 3NF         | Transitive dependencies        | "Do non-key attributes depend the key?"  |

# Table of Contents

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

107

122

Objective and Scope

Introduction to DBMS

Types of Databases

Database Languages

Basics of Relational Databases

Hands-On Tutorial on DQL

Database Design

Entity Relationship Model

Joins, Unions, and String Functions

Normalization

Motivation

Overview of Normal Forms

Real-Word Example

# Real-Word Example

## Initial (Unnormalized) Table

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Student_ID | Name<br>{Surname, First<br>Name} | Birthday<br>{DD.MM.YYYY} | Study_Program | Lectures<br>{Lecture_1; ...;<br>Lecture_n} |
|------------|----------------------------------|--------------------------|---------------|--------------------------------------------|
| 123456     | {Klein;<br>Stephanie}            | 01.05.2000               | BI            | {1. DBM; 2. ITM;<br>3. Stat}               |
| 123457     | {Schulze; Anton}                 | 09.11.2001               | PM            | {1. Stat; 2.<br>Math; 3. BWL}              |
| 123458     | {Schultze;<br>Stephanie}         | 01.03.2001               | PM            | {1. VWL}                                   |
| 123459     | {Bauer; Lisa}                    | 02.12.2000               | FI            | {1. Math; 2.<br>Stat}                      |

What is the problem with this data??

# Real-Word Example

## Initial (Unnormalized) Table

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

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**What is the problem with this data??**

Basic Questions to Ask ourself before we start!!!

- 1 Are there any Nulls ?
- 2 Is there anything missing?

108

122

# Real-Word Example

## First Normal Form (1NF)

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Student_ID | Name                | Birthday   | Study_Program | Lectures        |
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| 123458     | Schultze, Stephanie | 01.03.2001 | PM            | VWL             |
| 123459     | Bauer, Lisa         | 02.12.2000 | FI            | Math, Stat      |

- What is achieved in first normalization??
- Rule:: A relation is in First Normal Form (1NF) if all attributes contain only atomic (indivisible) values and there are no repeating groups.
  - ▶ No Nulls
  - ▶ No attribute with multiple values
  - ▶ No attribute with lists
  - ▶ Is ever value automatic?? – smallest possible division in terms of data
- Observe what do we have??



# Real-Word Example

## First Normal Form (1NF)

### Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

110

122

| Student_ID | Name                | Birthday   | Study_Program | Lectures        |
|------------|---------------------|------------|---------------|-----------------|
| 123456     | Klein, Stephanie    | 01.05.2000 | BI            | DBM, ITM, Stat  |
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- What is achieved in first normalization??
- Rule::
  - ▶ No Nulls
  - ▶ No attribute with multiple values
  - ▶ No attribute with lists
  - ▶ Is ever value automatic?? – smallest possible division in terms of data
- Observe what do we have??
- At first glance::
  - 1 No nulls → Yes. ✓
  - 2 Is every value atomic?? → No
  - 3 Does any attribute contain multiple values? → Yes
    - Name
    - Lectures
  - 4 Does any attribute contain lists or sets? → No ✓

# First Normal Form (1NF)

How to achieve?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Goal

Ensure that the table has atomic values and no repeating groups.

- **Aim:** Each field must contain exactly one value (Atomic value / Atomicity)
- Split composite attributes into smaller ones
  - ▶ Name → 1. first\_name ✓, 2. last\_name ✓
  - ▶ Horizontal Split
- What if we create new rows for Lectures ?
  - ▶ One Lecture each row
  - ▶ Vertical Split

# First Normal Form (1NF)

How to achieve?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Student_ID | First_Name | Last_Name | Birthday   | Study_Program | Lecture |
|------------|------------|-----------|------------|---------------|---------|
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            | DBM     |
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            | ITM     |
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            | Stat    |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            | Stat    |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            | Math    |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            | BWL     |
| 123458     | Stephanie  | Schultze  | 01.03.2001 | PM            | VWL     |
| 123459     | Lisa       | Bauer     | 02.12.2000 | FI            | Math    |
| 123459     | Lisa       | Bauer     | 02.12.2000 | FI            | Stat    |

- Is every value atomic?? **Yes ✓**
- No repeats?? **No**
- **Lots of redundancy**
- Can we define Primary keys tentatively?

# First Normal Form (1NF)

How to achieve?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

| Student_ID | First_Name | Last_Name | Birthday   | Study_Program | Lecture |
|------------|------------|-----------|------------|---------------|---------|
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            | DBM     |
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            | ITM     |
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            | Stat    |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            | Stat    |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            | Math    |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            | BWL     |
| 123458     | Stephanie  | Schultze  | 01.03.2001 | PM            | VWL     |
| 123459     | Lisa       | Bauer     | 02.12.2000 | FI            | Math    |
| 123459     | Lisa       | Bauer     | 02.12.2000 | FI            | Stat    |

- Is every value atomic?? **Yes ✓**
- No repeats?? **No**
- **Lots of redundancy**
- Can we define Primary keys tentatively?
  - ▶ Student\_ID
  - ▶ How about Study\_Program or say Lecture ?

# First Normal Form (1NF)

## Defining Keys

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Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Potential keys are:: Student\_ID    Lecture
- Why are they composite?
- A student can attend multiple lectures
- A lecture can be attended by multiple students
- The combination uniquely identifies each row

# Second Normal Form (1NF)

How to achieve?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Requirements

- 1 Data should be in 1NF
- 2 Reduce redundancy
- 3 No partial dependency on a composite key
- 4 Each row is uniquely identifiable

# Second Normal Form (1NF)

How to achieve?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Requirements

- 1 Data should be in 1NF
- 2 Reduce redundancy
- 3 No partial dependency on a composite key
- 4 Each row is uniquely identifiable

What we have??

- Data should be in 1NF **Yes ✓**
- Redundancy → **Yes we have it**
- No partial dependency on a composite key **Yes we have it**

# Second Normal Form (2NF)

What is partial dependency?

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- **Step 1:** Identify the primary key → **StudentID** ✓
  - ▶ Is it composite (more than one column)? **No** ✓
- **Step 2:** Check dependencies
  - ▶ Ask for each non-key attribute: “Does this attribute depend on the entire key, or only part of it?”
  - ▶ (Student\_ID, Lecture) → Study\_Program
  - ▶ Student\_ID → Study\_Program
  - ▶ This means Lecture depends on Study\_Program and Student\_ID
- **Step 3:** Ensure each row is uniquely identifiable

**Solution: Try splitting the data in different tables!**



# Achieving the Second Normal Form (2NF)

## Splitting the data in two tables

Database Systems  
and Information  
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Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

116

122

Table: 1. Student Enrollment

| Student_ID | First_Name | Last_Name | Birthday   | Study_Program |
|------------|------------|-----------|------------|---------------|
| 123456     | Stephanie  | Klein     | 01.05.2000 | BI            |
| 123457     | Anton      | Schulze   | 09.11.2001 | PM            |
| 123458     | Stephanie  | Schultze  | 01.03.2001 | PM            |
| 123459     | Lisa       | Bauer     | 02.12.2000 | FI            |

Table: 2. Lectures

| Student_ID | Lecture |
|------------|---------|
| 123456     | DBM     |
| 123456     | ITM     |
| 123456     | Stat    |
| 123457     | Stat    |
| 123457     | Math    |
| 123457     | BWL     |
| 123458     | VWL     |
| 123459     | Math    |
| 123459     | Stat    |

No Partial Dependencies ✓

# Third Normal Form (3NF)

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Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

## Definition

A relation is in Third Normal Form (3NF) if it is in 2NF and no non-key attribute is transitively dependent on the primary key. **Rule:**

- Must already be in 2NF
- No transitive dependencies
- Non-key attributes must depend only on the primary key

What we have??

- Must already be in 2NF **Yes ✓**
- Non-key attributes must depend only on the primary key **Yes ✓**
- No transitive dependencies **Yes ✓**
- The table is already in 3NF

# Third Normal Form (3NF)

## Hypothetical dependency

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Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

### Transitive Dependency ::

- Student\_ID determines Study Program
- Study Program determines program details
- This creates a potential transitive dependency:

*Student\_ID → Program → Program\_Name*

# 3NF Decomposition

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Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Two more tables can be added to remove this dependency

Table: 3.Program Details

| Program_ID | Program_Name         |
|------------|----------------------|
| BI         | Business Informatics |
| PM         | Project Management   |
| FI         | Finance              |

Table: 4.Lecture Details

| Lecture_ID | Lecture_Name            |
|------------|-------------------------|
| DBM        | Database Management     |
| ITM        | IT Management           |
| Stat       | Statistics              |
| Math       | Mathematics             |
| BWL        | Business Administration |
| VWL        | Economics               |

119

● Primary Key : Student\_ID

122

● Foreign Keys : Program\_ID    Lecture\_ID

# Final Result and Benefits

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- Each table represents one entity
- No redundancy
- No anomalies
- Referential integrity possible
- Database is in Third Normal Form (3NF)

120

122

# Key Exam Takeaways

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

- 1NF: atomic values
- 2NF: no partial dependency
- 3NF: no transitive dependency
- Always identify functional dependencies first

121

122

## Database Systems and Information Integration

Dr. Rushikesh  
Amrutsamanvar

Objective and  
Scope

Introduction

Introduction to  
DBMS

Types of  
Databases

Database  
Languages

Basics of  
Relational  
Databases

Hands-On Tutorial  
on DQL

Database Design

Entity Relationship  
Model

Joins, Unions, and  
String Functions

Normalization

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Thank you!